

Computational Control Rocket Landing Project

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June 2026

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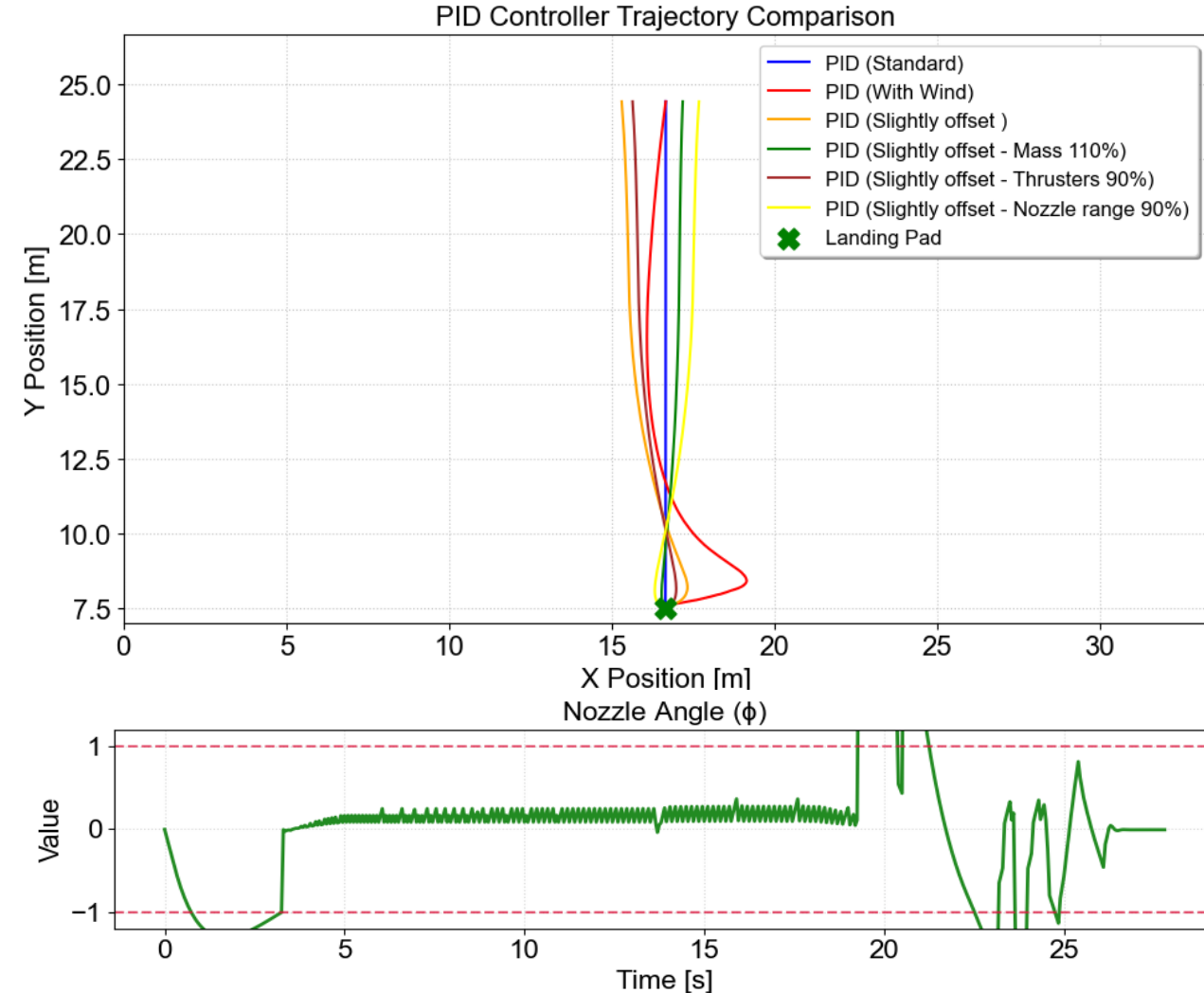
Current state

The existing PID controller was tested to evaluate the current performance and establish a baseline.

The controller successfully achieves rocket landing under nominal conditions – centered initial position, standard thruster efficiency, nozzle angle range and mass.

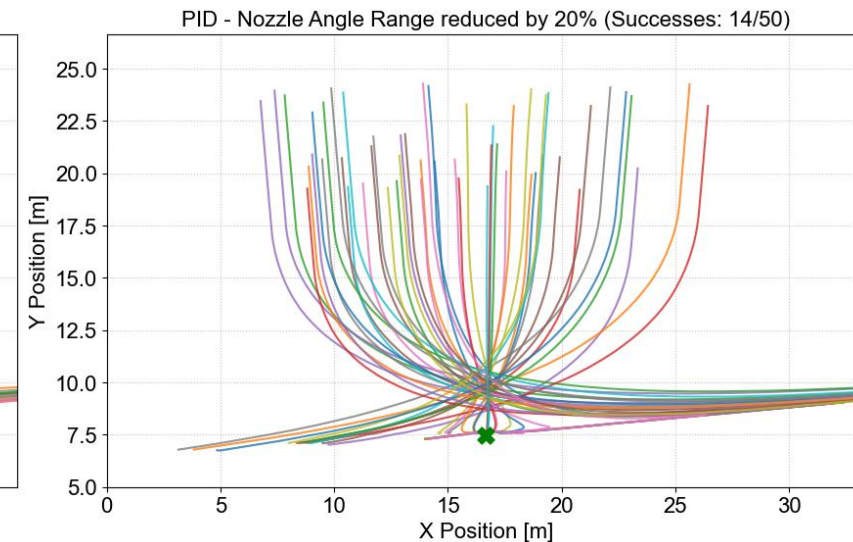
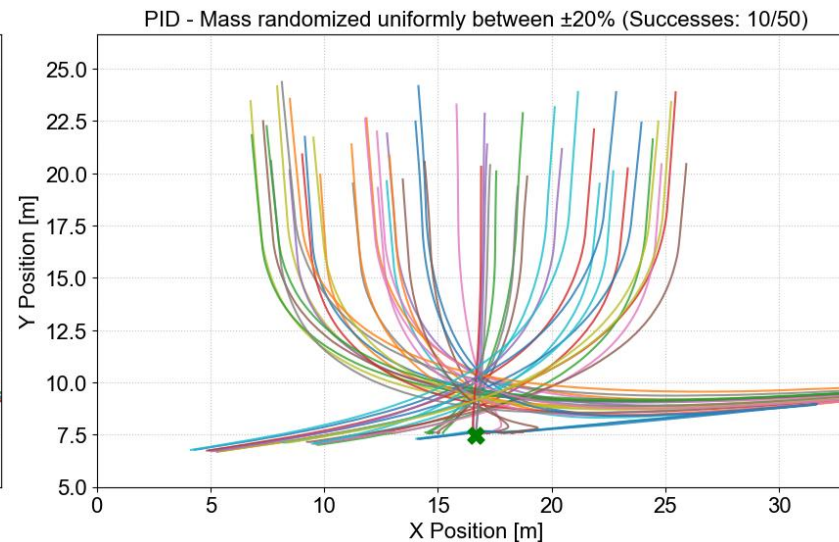
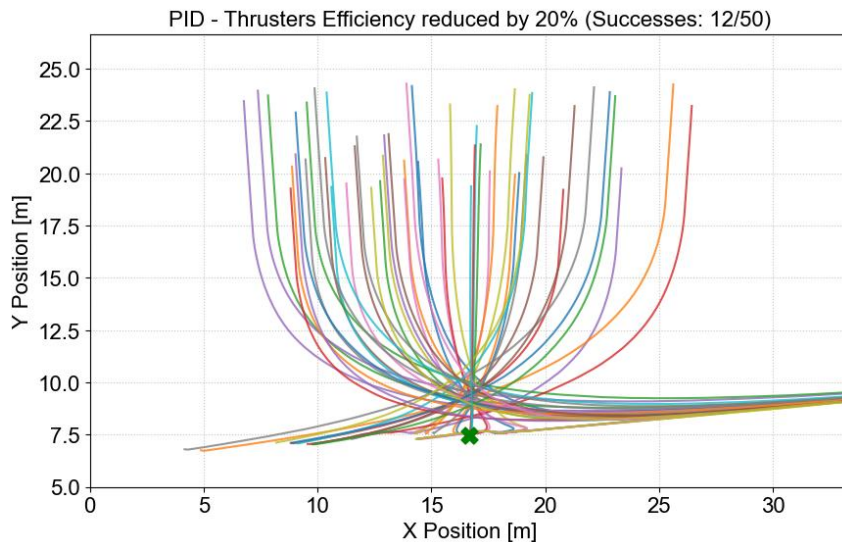
While the PID controller accommodates small deviations from the aforementioned nominal conditions, it fails to recover from larger deviations.

A key limitation of the PID controller is its lack of constraint awareness: it might command control inputs that exceed physical system limits.



Failure modes

To evaluate the robustness to malfunctions, for each malfunction the PID controller was tested on 50 random initial position rollouts, accounting for real-world positioning and navigation uncertainties. Realistic test cases, such as a 20% deviation from the nominal case, were examined **individually** to verify the performance, but the PID failed to guarantee satisfactory performance. Given the baseline system's inability to withstand moderate deviations, evaluating more severe test cases was unnecessary. These limitations establish a clear imperative for a more robust control architecture.



PID controller is not robust to malfunctions

24%
Success rate

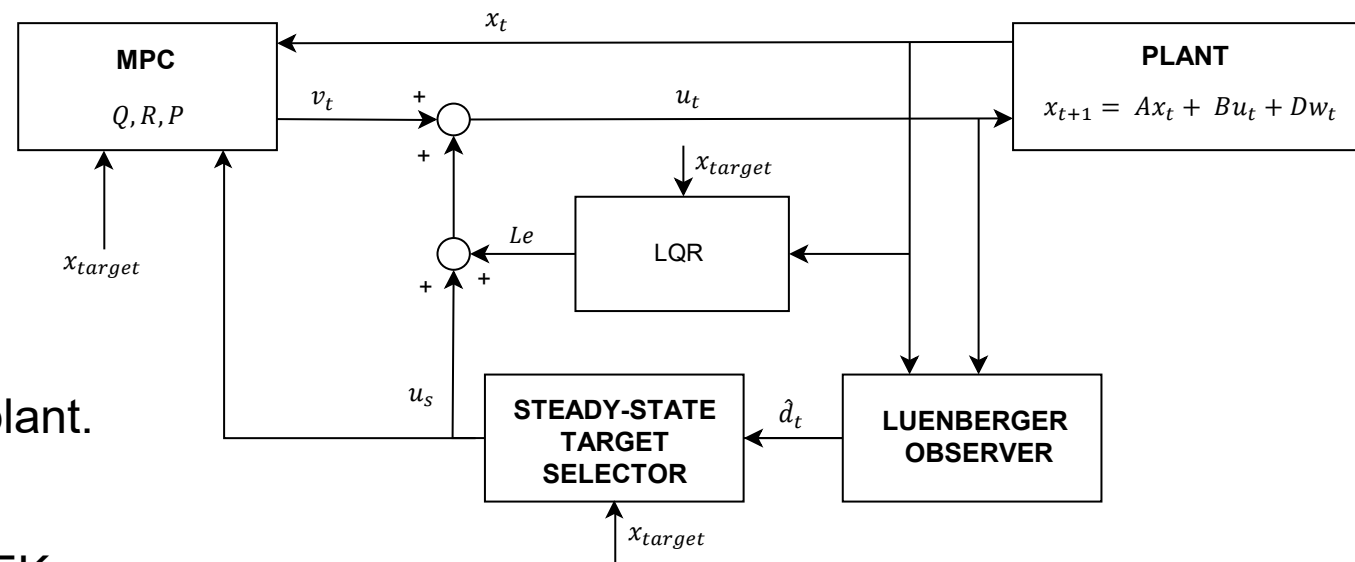
MPC + Disturbance Estimation + Nested LQR

Given that the problem comes with a model of the plant, constraints and a target state to track, MPC constitutes an ideal framework. To counter potential model mismatches and environmental disturbances, robustness is significantly enhanced through two additional layers.

- **Luenberger observer:** estimates disturbances (from the mismatch between prediction and measurement) and corrects the steady-state target,
- **Nested LQR controller:** rejects fast disturbances, specifically prioritizing orientation.

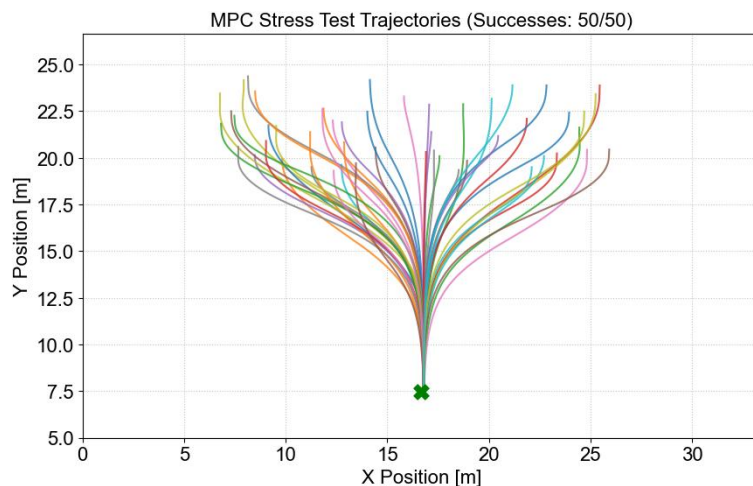
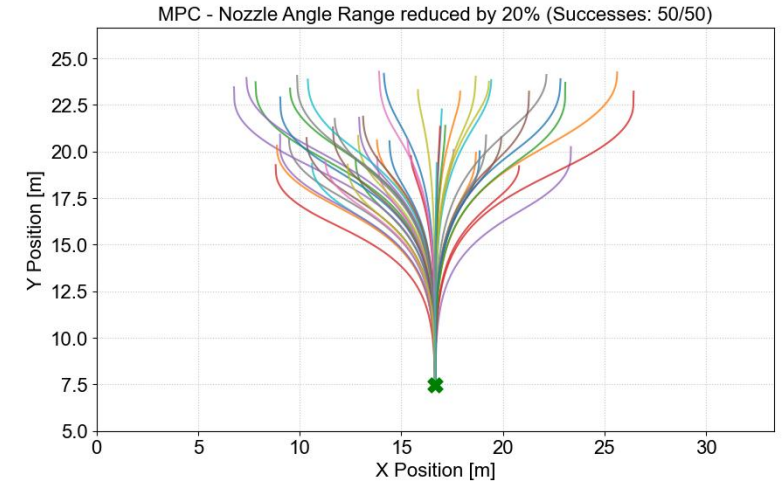
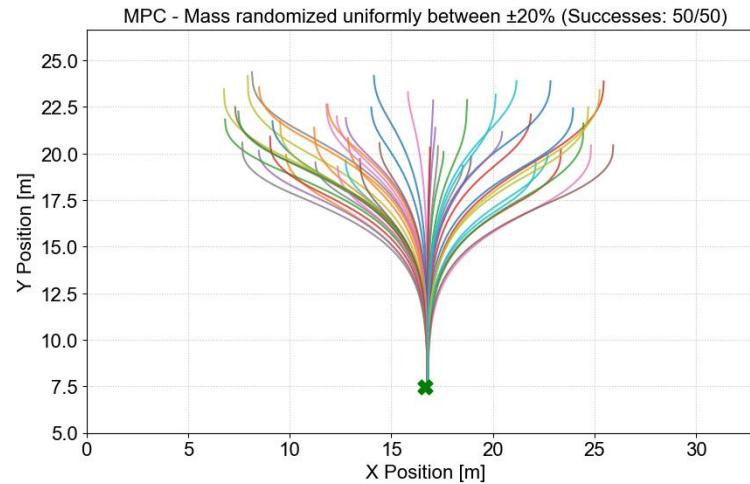
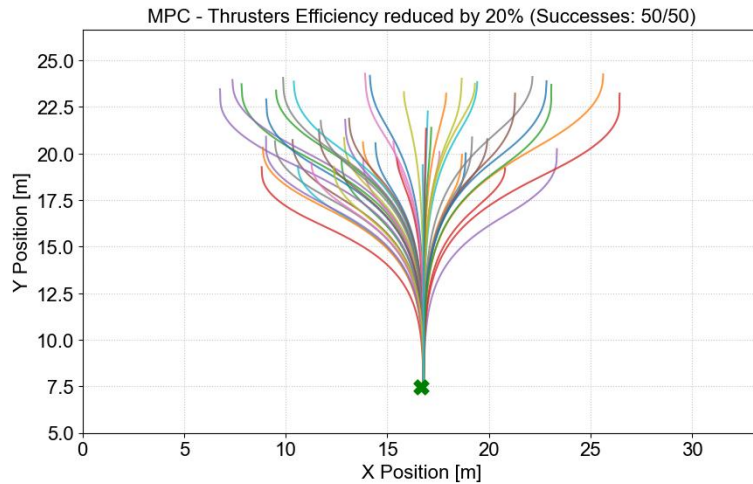
Main strengths:

- Predicts future trajectories and anticipates the effect of inputs by leveraging the **model** of the plant.
- Handles **constraints** by solving a constrained convex optimization problem online using MOSEK.
- Achieves **robustness to disturbances** by estimating disturbances through the Luenberger observer, correcting the steady-state target and applying stabilizing feedback through the nested LQR.



Demonstrations

Tested on the same malfunctions, the proposed controller completely neutralizes the vulnerabilities of the PID.



To prove the enhanced robustness of the new controller, extreme stress tests were conducted: these evaluations **combined** critical 45% reduction in (main and side) thruster efficiency and nozzle angle range, $\pm 50\%$ mass variations, and randomized initial states. The controller seamlessly mitigated these compounded degradations, achieving a 100% mission success rate across all test episodes.

✓ MPC succeeds in all episodes as well as stress tests!

100%
Success rate

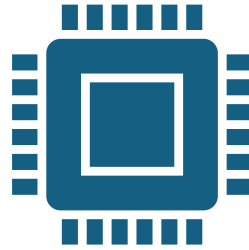
Challenges

Despite its strong performance, some aspects of the recommended controller must be verified during testing.



Model accuracy

This MPC implementation builds upon a LTI model: if the model is inadequate (i.e. the LTI assumption is not valid because of large deviations from the linearization condition - hovering) the performance might degrade. Online linearization could help.



Computational cost

Even though MOSEK is an efficient state of the art solver, it is necessary to verify whether the computing power available on the rocket is enough to run the controller in real time during deployment.



Stability

During testing and development, it is necessary to verify the Time Horizon $\tilde{N}$ such that if $N > \tilde{N}$ closed-loop stability is guaranteed. Although the chosen horizon demonstrates empirical sufficiency, the theoretical lower bound $\tilde{N}$ has not been verified rigorously.